Praise Worthy Prize

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Sizing of Optimal Case of Standalone Hybrid Power System Using Homer Software

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Abstract – *In rural areas, the high cost of grid extension requires institutions to consider other alternatives, such use of generators (GE) diesel is often considered as an economic and reliable solution, but generates environment pollution and it is convenient for the user. High Costs of operation; Energy dependence; fuel supply problem (removal of fuel suppliers); complicated and expensive maintenance; Low lifetime (about 5 years and sometimes less); Impossibility of 24 hours electricity production (or need for several GE; Sound Births and waste oil management issues). Furthermore, the continuous decline in the prices of generators based on renewable energy and the increasing reliability of these systems lead to a greater use of renewable energy sources for power generation in remote areas. A property which limits the use of renewable energy is related to the variability of resources. Fluctuations in load according to annual or daily periods are not necessarily correlated to the resources. In remote areas, the preferred option is the coupling between multiple sources, such as wind turbines and solar panels, this coupling is called hybrid power system. Algeria's geographic location presents* several advantages for the development and use of renewable energy, namely, solar energy and wind energy. In addition, Algeria has huge deposits of natural gas, 98% electricity comes from gas. Therefore, currently, the production of electricity from renewable energies primarily depends on their competitiveness with economic gas.The objective of this work is to study the technological feasibility and economic viability of the hybrid system (PV/fuel)electrification project in a school located in Tarat (Algeria) and to reduce traditional power emissions by using renewable energy. The HOMER model is used in this study to size the proposed system and determine the optimum configuration.*

***Keywords**: Solar Energy, Hybrid System, Homer, Photovoltaic System*

Nomenclature

GE	Generator
PV	Photovoltaic
HOMER	Hybrid Optimization of Multiple Electric Renewables
DC	Direct Current
AC	Alternating current
C_{NPC}	The Net present cost of system
$C_{ann,tot}$	The total annualized [$/yr]
i	The annual interest rate [%]
N	Number of years
R_{proj}	The project life time [yr]
$CRF(i,N)$	The capital recovery factor
COE	The levelized cost of energy
E_{prim}	The total amount of primary load [kWh/yr]
n_r	The reference module efficiency,
T_{cref}	The reference cell temperature in degree Celsius
$G(t)$	The solar irradiation in tilted module plane [Wh/m^2]
B	The temperature coefficient
T_c	The cell effective temperature
A	The PV generator area [m^2]
E_{gen}	The generator's total annual electrical production [kWh/yr]
m_{fuel}	The generator's total annual fuel consumption [kg/yr]
LHV_{fuel}	The lower heating value of the fuel [MJ/kg]
η_{gen}	The average electrical efficiency of the generator over the year[%]
$\eta gen,tot$	The average total efficiency of the generator over the year[%]
F	Fuel consumption this hour [L]
$F0$	Generator fuel curve intercept coefficient [L/hr/kWrated]
$F1$	Generator fuel curve slope[L/hr/kW]
Ygen	Rated capacity of the Generator [kW]
Pgen	Output of the generator in this hour [kW]
Crep,gen	Generator replacement cost
Rgen,h	Generator lifetime[hr]
Ngen	The number of hours the generator operates during one year [hr/yr]
Ftot	Total annual generator fuel consumption [L/yr, m^3/yr, or kg/yr]

I. Introduction

Developing a hybrid power system for any application such as homes, educational institutes, industries; an engineer needs a basic knowledge of technology, availability of resources, economics, sizing, regulations. Renewable energy is the energy that comes from resources naturally replenished on a human timescale, such as sunlight, wind, rain, tides, waves, and geothermal heat. It replaces conventional fuels in four distinct areas: electricity generation, air and water heating/cooling, motor fuels, and rural (off-grid) energy services. To generate electric power with good efficiency, always co-generation or hybrid systems are always useful. A hybrid energy system usually consists of two or more renewable energy sources used together with conventional resources to provide more efficiency to the system and greater balance in energy supply to load [1].

The intermittent nature of Solar and wind leads to power outages due to insufficient solar energy and lack of wind speed. Hence diesel generator is also considered for its continuous electrical power generation [2]. To ensure the reliable operation of the system, a storage device can be incorporated to meet the load demands. Researchers proved that the hybrid PV/Wind/battery system is a reliable electricity source for which public support is globally increasing day by day [3],[4].

The energy generated by people can be utilized in their homes and the excess can be sent to the grid, which will be an additional income[5].In order to develop a hybrid power system model to identify the economic configuration of PV and Wind with Diesel generation, the National Renewable Energy Laboratory's Hybrid Optimization of Multiple Electric Renewables (HOMER) is used. HOMER Pro is used to design and find optimized configurations of a hybrid power system in terms of stability, low cost, size and number of components, prior to installation [6],[7].

The aim of this paper is to design a hybrid power system model with metrological data inputs, also to describe feasible technology, component costs based on the availability of resources. The data considered as inputs is used to simulate different system configurations, or combinations of components, and generate results viewed as a list of feasible configurations sorted by net present cost [8].

The Present work proposes the simulation of an hybrid power system, with an economic combination PV, Diesel Generator, Converter and battery storage [9] – [10]. The simulated results using HOMER give the comparative economic analysis with each configuration and evaluate the best configuration. This is evident in the results presented in this paper.

Thus HOMER is advantageous in performing energy balance configuration for each hour to choose whether the configuration is feasible or not. For the same feasible combination the total cost of operation is estimated over the life time of the project in a particular area before installation [11].

II. State of Art of Hybrid Power Systems

In recent years there is a great effort for advancement in power generation considering environment friendly technologies such as renewable energy technologies is becoming, as technically viable and environmentally friendly. Around the world, effort is being made to study the viability of renewable energy incorporated hybrid systems as alternatives to diesel generator [12], [13].

The hybrid energy system generally works with a primary renewable source in parallel combination with a standby secondary non-renewable module and a storage unit. Hybrid energy systems offer an attractive and practical solution to meet the electrical power demand in rural communities around the globe [14], [15].

The Main drawbacks of the standalone system considering both solar and wind energy is energy fluctuation, resulting in intermittent delivery of power and causing problems if reliable and continuous supply is required. These problems can be overcome with the use of standalone hybrid systems.

A hybrid power system can be defined as a combination of different energy sources, but complementary energy generation system can be provided by renewable or mixed (RES- with a backup of diesel generator set) energy. Hybrid systems provides better features of each energy resource compared to conventional sources and also provide “grid-quality”electricity, with a power range up to several hundred kilowatts. The major advantage of a hybrid system is that when one power source is at low levels the other source is usually at higher levels.

A solar panel is less effective on cloudy and windy days so it will produce lower energy levels while the wind generator may be produce a larger amount of energy. Similarly, for wind generation the main issue is the location of the site, which has a certain amount of wind on regular basis. The major use of non-conventional energy makes this system almost independent and lowers energy prices over the long-term, and diesel generator combination is used as back up in emergency situations such as high loads or low renewable power availability.

III. Geographical Description of the Village

Tarât area (Fig. 1) is known for the largest gas field in Algeria. Geographically, it lies on latitude 26 ° 07 North, and longitude 9 ° 21 is.

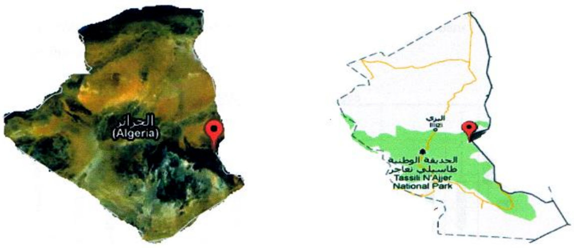


Fig. 1. Geographical situationof "Tarât"

It has a continental and semi-arid climate with very high temperatures in summer, very cold in winter and with very low rainfall. In addition to its gas field, the Tarât region also has a solar field among the largest in Algeria.

III.1. Solar Potential

Tarât has a great solar potential, the monthly average radiation varies between 4 and 8 kWh/m^2 as presented in Figure 2.

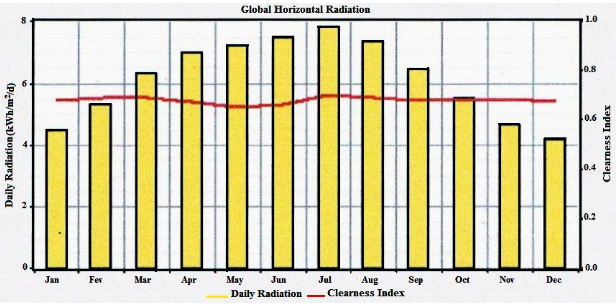


Fig. 2. Monthly average solar irradiation

A significant solar potential can be noticed in summer months with a maximum during the month of July and a minimum in winter during the month of December.

III.2. Wind Potential

Tarât is considered as a low windy area in Algeria, the average annual speed of 4.2 m /s, as shown in Figure 3.

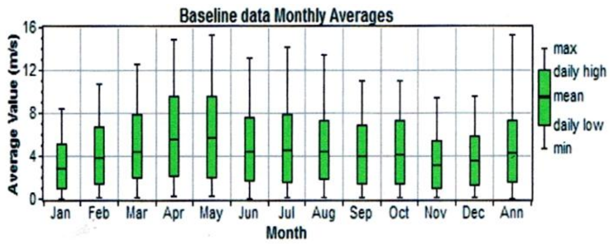


Fig. 3. Monthly wind speed evolution

IV. Hybrid System PV/Generator

This system uses the advantages of photovoltaic and diesel, propane or gasoline generators. The photovoltaic system provides intermittent energy which is often less expensive in remote areas.

The generator uses energy, depending on the demand. Thistype of system is particularly applicable to remote locations where it is important to have electricity at any time, or where fuel transportation costs are high and where it is not yet profitable to use only the photovoltaic system with batteries. Hybrid Systems PV / Generators are often used in communications towers as well as in shelters and logging camps in remote areas. They can also be associated to other energy sources such as wind turbines and hydro plants, when there is a complementarity of electrical production.

The school electrical needs are very important related to other elements, therefore an hybrid system was proposed. The average consumption of various school facilities has been estimated at around 6.4 kWh/d. based on technical data in Table I, it is possible to have an estimate of the school consumption profile in Tarât, as shown in Figure 4.

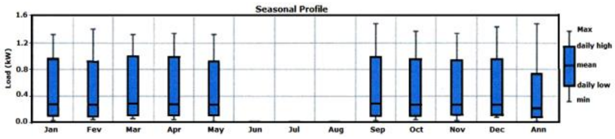


Fig. 4. Seasonal load profile

TABLE I
SCHOOL CONSUMPTION

Apparatus feed	Power (W)	Numbers elements	Duration of run	Total consumption
Lamp	14	5	6	420
Kitchen lamp	20	4	5	400
Lamp direction	20	2	5	200
Magazin	20	4	3	240
Fridge	300	1	13	3900
Canteen lamp	20	6	3	360
Printer	500	1	1	500
microcomputer	130	1	5	650
Total consumption				6670

IV.1. Optimizing Hybrid System for School

The system selects from data and the Homer software displays the following schema.

The proposed model of standalone hybrid power system, shown in Fig. 5, considering the metrological data of "Tarat" campus as shown in Fig. 1. Metrological data inputs serve to analyze the system performance. The proposed model is designed with various inputs describing technology options, availability of resources and component costs in a prior mentioned area.

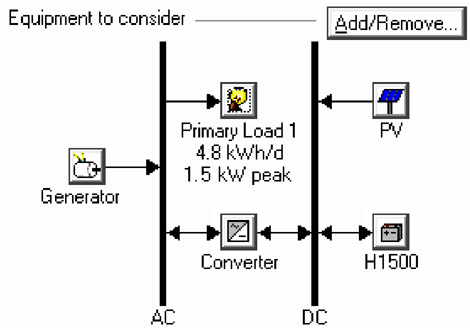


Fig. 5. Proposed standalone hybrid power system model

IV.2. Electrical Charge

The electrical charges make electrical power useful. Resistive expenses include incandescent bulbs, water heaters, etc. Devices using electrical machines are loaded by resistances and inductances. They are the main consumers of reactive power.

Accrued DC may also have inductive components, but only the effects introduced by these are transient voltage and current variations during changes in the system functioning.

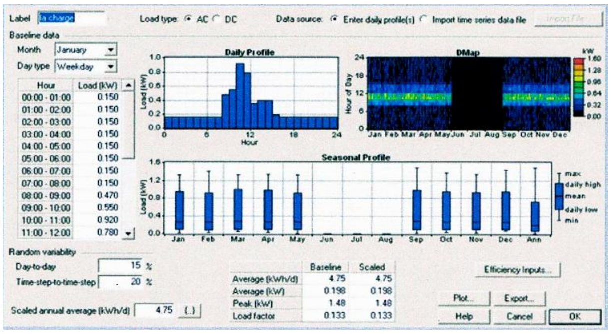


Fig. 6. Load profile

IV.3. HOMER Solftware

The micro power optimization model simplifies the task of evaluating the designs of both off-grid and grid-connected power systems for a variety of applications. When a power system is designed, many decisions must be taken about the configuration of the system: Which components is it reasonable to include in the system design? How many components and of which size should be used? The large number of technology options and the variation in technology costs and availability of energy resources make these decisions difficult. HOMER's optimization and sensitivity analysis algorithms make it easier to evaluate the many possible system configurations.

To use HOMER, the model is provided with inputs, which describe technology options, component costs, and resource availability. HOMER uses these inputs to simulate different system configurations, or combinations of components, and generates results that that can be seen as a list of feasible configurations sorted by net present cost. HOMER also displays simulation results in a wide variety of tables and graphs that help to compare configurations and evaluate them on economic and technical basis. The tables and graphs for use in reports and presentations can be exported.

When the need is to explore the effect that changes in factors such as resource availability and economic conditions might have on the cost-effectiveness of different system configurations, it is possible to use the model to perform sensitivity analyses. To perform a sensitivity analysis. To do that, HOMER is provided with sensitivity values that describe a range of resource availability and component costs.

HOMER simulates each system configuration over the range of values. The results of a sensitivity analysis can be used to identify the factors that have the greatest impact on the design and operation of a power system. HOMER sensitivity analysis results can also be used to answer general questions about technology options to inform planning and policy decisions.

HOMER simulates the operation of a system by making energy balance calculations for each of the 8,760 hours in a year. For each hour, HOMER compares the electric and thermal demand in one hour to the energy that the system can supply in that hour, and calculates the flows of energy to and from each component of the system. For systems that include batteries or fuel-powered generators, HOMER also decides for each hour how to use the generators and whether to charge or discharge the batteries. HOMER performs these energy balance calculations for each system configuration to consider. It then determines whether a configuration is feasible, i.e., whether it can meet the electric demand under the conditions specified, and estimates the cost of installing and using the system over the project lifetime.

The system cost calculations account for costs such as capital, replacement, operation and maintenance, fuel, and interest. After simulating all of the possible system configurations, HOMER displays a list of configurations, sorted by net present cost (sometimes called lifecycle cost), that can be used to compare system design options.

When sensitivity variables are defined as inputs, HOMER repeats the optimization process for each sensitivity variable specified. For example, if wind speed is defined as a sensitivity variable, HOMER will simulate system configurations for the range of wind speeds specified.

IV.4. Analytical Model of Solar Energy Generation

According to previous works done by[16] [17] [18] [19] the model of power generated by a PV modulecan be given by the below formula:

$$P(t) = n_r\left[1-\beta\cdot\left(T_c-T_{cref}\right)\right]\cdot A\cdot G(t) \qquad (1)$$

A "generator" is a device that consumes fuel to produce electric (and sometimes thermal) energy. Generators can be dispatched, meaning that the system can turn them on when necessary. Microturbines and fuel cells are generators, as are diesel- and gasoline-fueled reciprocating engine generators.

This is the average electrical efficiency of the generator over the year, defined as the electrical energy out divided by fuel energy in. HOMER uses the following equation to calculate the average electrical efficiency:

$$\eta_{gem} = \frac{3.6\ \cdot E_{gem}}{m_{fuel}\ \cdot\ LHV_{fuel}} \qquad (2)$$

The factor of 3.6 in the above equation arises because 1 kWh = 3.6 MJ. This is the average total efficiency of the generator over the year, defined as the electrical plus thermal energy out divided by fuel energy in. HOMER uses the following equation to calculate the average total efficiency:

$$\eta_{gem} = \frac{3.6 \cdot (E_{gem} + H_{gem})}{m_{fuel} \cdot LHV_{fuel}} \quad (3)$$

IV.5. Generator Carbon Monoxide Emissions Factor

The Fuel that the generator consumes; the amount of carbon monoxide emitted per unit. Because carbon monoxide is a product of incomplete combustion, the produced quantity will depend on the fuel, engine design, and operating conditions, including the power output of the generator. But HOMER makes a simplifying assumption that this factor is constant.

The following graph shows the value of the carbon monoxide emissions factor for diesel generators in the 50 kW - 450 kW size range. The source of these data is an internal NREL report by Erin Kassoy entitled "Modeling diesel exhaust emissions in diesel retrofits". The default value for the generator CO emissions factor is equal to the average value between 50% and 100% load.

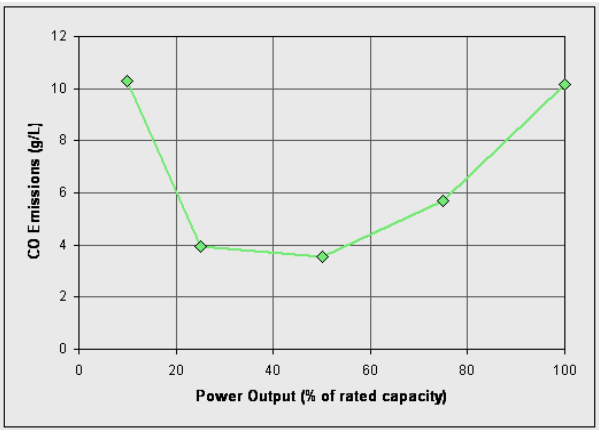


Fig. 7. Value of the carbon monoxide emissions factor for diesel generators

IV.6. Generator Derating Factor

The Maximum power of a generator operating at the minimum fossil fraction, as a percentage of its rated output.For example, a 20 kW diesel generator is modified to run on a mixture of diesel fuel and biogas, with a minimum diesel percentage of 20%. If the output of the engine is limited to 15 kW when operating at 20% diesel fraction, the derating factor would be 15 kW divided by 20 kW, or 75%. HOMER assumes that a cofired generator can produce up to 100% of its rated output, provided that the fossil fraction is high enough. In the above example, the generator could produce up to 20 kW, but the diesel fraction would have to exceed 20% for any output power above 15 kW.

IV.7. Generator Fuel Cost

The Annual cost of generator supply. HOMER calculates this value by multiplying the fuel price by the amount of fuel used by the generator in one year.

If the generator burns biogas, either as its primary fuel or cofired with another fuel, HOMER includes the biomass cost in the generator fuel cost. The biomass cost is equal to the amount of biomass *feedstock* consumed over a year multiplied by the price of biomass.

IV.8. Generator Fuel Curve Intercept Coefficient

The fuel curve intercept coefficient is the no-load fuel consumption of the generator divided by its rated capacity. If you were to plot a straight line of fuel consumption versus the power output of the generator, the y-intercept of that line divided by the generator size is equal to the fuel curve intercept coefficient.

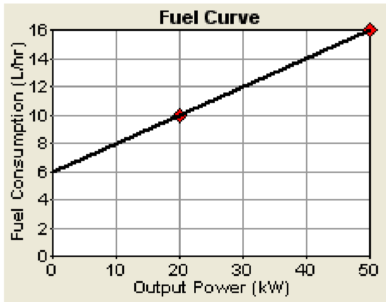


Fig. 8. Fuel curve intercept coefficient

If the generator is running in a particular hour, HOMER calculates the fuel consumption for that hour using the following equation:

$$F = F_0 \cdot Y_{gem} + F_1 \cdot P_{gem} \quad (4)$$

If the generator is not running in a particular hour, then the fuel consumption for that hour is zero.

IV.9. Generator Fuel Curve Slope

The fuel curve slope is the marginal fuel consumption of the generator, in units of fuel per hour per kW of output. If you were to plot a straight line of fuel consumption versus the power output of the generator, the slope of that line is the fuel curve slope, the same as Figure 8.

IV.10. Generator Hourly Replacement Cost

The generator lifetime is specified in operating hours number. So the hourly replacement cost of each generator can be calculated according to the following equation:

$$C_{rep.gem} = \frac{C_{rep.gen}}{R_{gen,h}} \quad (5)$$

IV.11. Generator Operational Life

In HOMER, the lifetime of generators is specified in terms of operating hours. The number of years that a generator will last is therefore an output variable, which HOMER calculates according to the following equation:

$$R_{gen} = \frac{R_{gen,h}}{N_{gen}} \tag{6}$$

For example, if the generator has a lifetime of 20,000 operating hours and HOMER determines that it will operate 4300 hours per year, then its expected lifetime in years would be 20,000 hours / 4300 hours per year = 4.65 years.

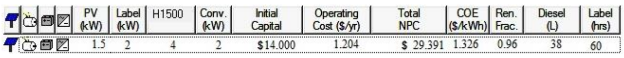
IV.12. Specific Fuel Consumption

The specific fuel consumption is the average amount of fuel consumed by the generator per kWh of electricity it generates. HOMER calculates the specific fuel consumption using the following equation:

$$F_{spec} = \frac{F_{tot}}{E_{gen}} \tag{7}$$

IV.13. The Optimal Solution Given by HOMER

Fig. 9 shows the optimal solution that contains the investment cost.



				PV (kW)	Label (kW)	H1500	Conv. (kW)	Initial Capital	Operating Cost ($/yr)	Total NPC	COE ($/kWh)	Ren. Frac.	Diesel (L)	Label (hrs)
				1.5	2	4	2	$14.000	1.204	$ 29.391	1.326	0.96	38	60

Fig. 9. Optimum Solution choosen by HOMER

HOMER software gives the following configuration as the optimum solution:

- Photovoltaic solar panel with global power of 1.5kVA.
- Fuel generator with power of 2 kVA.
- 4 batteries with stored energy of 1500Ah and 48 volt.
- Energyconverter of 2 kVA.

TABLE II
PERCENTAGE OF ELECTRICITY HYBRID SYSTEMS

Production	kWh/yr	%
PV array	2.741	96
Generator 1	112	4
Total	2.853	100

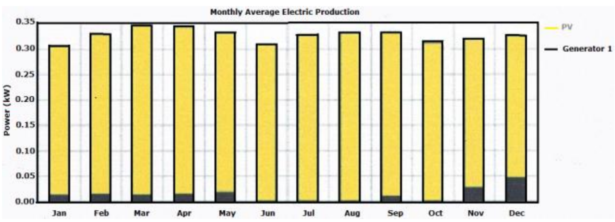


Fig. 10. Profile of energy production

From these results, it is possible to note that 96% of the school energy needs is provided by the PV generators, the remaining 4% being provided by the diesel generator. This fraction is 96% satisfactory, keeping into account the available solar resource.

IV.14. System Emissions

It can be noted that the supply with a large percentage of PV production reduces emissions of greenhouse gases and thus lowers pollution.

TABLE III
EMISSIONS OF HYBRID SYSTEMS

Polluant	Description	Emissions (kg/yr)
Carbon dioxide (CO2)	Non toxic greenhouse gas	98.8
Carbon Monoxide (CO)	Poisonous gas produced by incomplete burning of carbon in fuels. Prevents delivery of oxygen to the body's organs and tissues, causing headaches, dizziness, and impairment of visual perception, manual dexterity, and learning ability.	0.244
Unburned Hydrocarbons (UHC)	Products of incomplete combustion of hydrocarbon fuel, including formaldehyde and alkenes. Lead to atmospheric reactions causing photochemical smog.	0.027
Particulate Matter (PM)	A mixture of smoke, soot, and liquid droplets that can cause respiratory problems and form atmospheric haze.	0.0184
Sulfur Dioxide (SO_2)	A corrosive gas released by the burning of fuels containing sulfur (like coal, oil and diesel fuel). Cause respiratory problems, acid rain, and atmospheric haze.	0.198
Nitrogen Oxides (NO_x)	Various nitrogen compounds like nitrogen dioxide (NO_2) and nitric oxide (NO) formed when any fuel is burned at high temperature. These compounds lead to respiratory problems, smog, and acid rain.	2.18

IV.15. Photovoltaic System Production Profile

Figure 11 shows that the photovoltaic production depends on solar potential and influence of the ambient temperature with a peak production of around 12 hours.

IV.16. Production Profile of Diesel Generator

The diesel generator is not made within the month of October (Fig. 12). In addition, the group operates at its minimum output hang the other months of the year and this because of the high photovoltaic production (Figure 10).

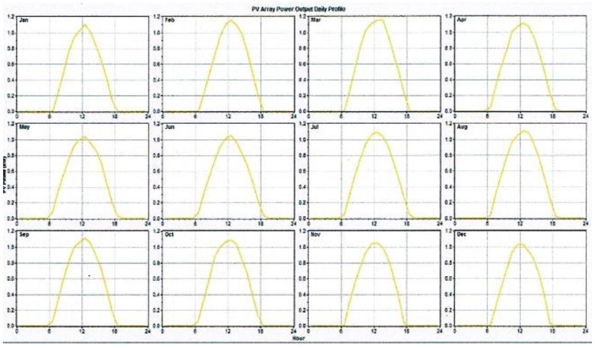


Fig. 11. PV production profile in one year

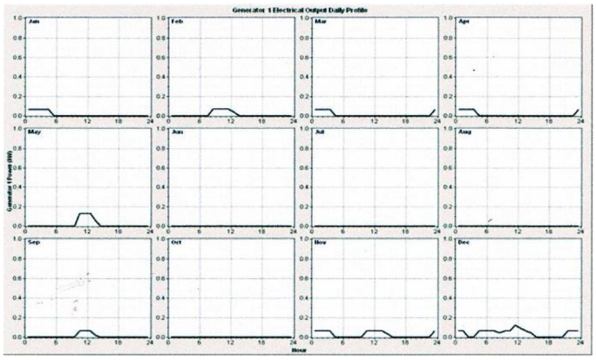


Fig. 12. Yearly production profile for diesel generator

IV.17. Charge Profile

Figure 13 shows that the load curve varies according to the load demand, therefore there is a consumption peak from 10 am to 11 am.

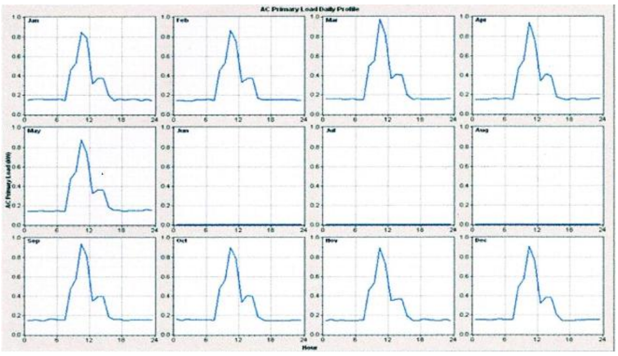


Fig. 13. Yearly charge profile

Converters:

In hybrid systems, converters are used to charge storage batteries and to transform the DC /AC. Three types of converters are often found in hybrid systems: rectifiers, inverters and choppers.

The rectifiers perform AC / DC conversion. In the hybrid system, they are often used to charge batteries from a source AC source. These are relatively simple devices, cheap and good yield. The inverters convert DC to AC They can autonomously operate for supplying loads at AC. The choppers, the third type of converters, are used to perform the DC / DC conversion, for example, to adjust the tension between two sources.

UPS conversion profile:

This figure shows that the converter operates in inverter mode to confirm photovoltaic production.

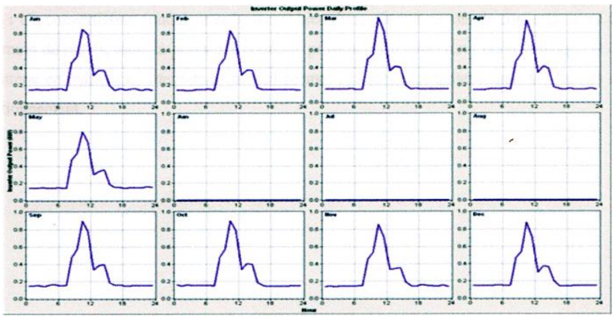


Fig. 14. Conversion profile annual inverter mode

Rectifier conversion profile:

A converter (inverter-rectifier) converts well, with minimum losses, the whole continuous power from the photovoltaic generator and battery power as an alternative, and allows charging battery in its operating limits.

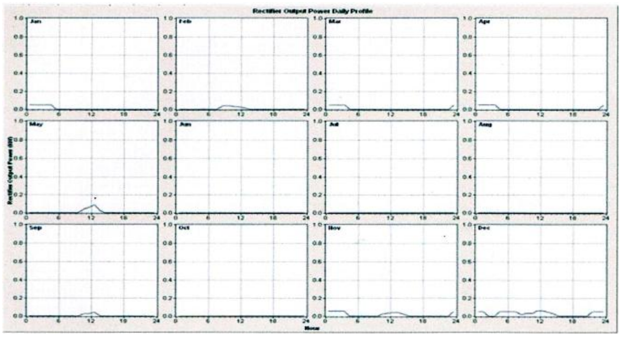


Fig. 15. Conversion profile annual rectifier mode

V. Role and Characteristics of Battery Storage

The function of the accumulator in a photovoltaic system is twofold.

First, the system reserves or stores energy, secondly, it has an impedance matching function. The photovoltaic generator does not fix system voltage. To fix and optimize the operating point of the modules, it is necessary that part a of the system fixes the voltage of the generator. Batteries are able to play this role because their voltage varies depending on conditions.

The specific character of photovoltaic (relatively expensive energy) installed-alone imposes the choice of batteries with the following characteristics:

- Load Performance /High discharge (low internal resistance, low self-discharge losses);
- Good ability to cycling (good life);
- Large electrolyte reserve;

- Cost as low as possible.

The storage system is often used in small hybrid systems ending power supplies for a relatively long time (hours or days). It is used to eliminate power fluctuations to current terms.

Energy storage is generally achieved through the use of batteries. The most commonly used batteries are lead acid batteries (Pb-acid) and batteries with nickel-cadmiun (Ni-Cd). Each has its own merits and according to construction methods, they have very different performance characteristics. The lead acid battery is the best known, being used for over 15 years to deliver the cars starting current, electricity, emergency systems and electric vehicles traction.

V.1. Load Profile and Battery Park Landfill

The charging process and discharging stands between 6 am and 18:00 pm.

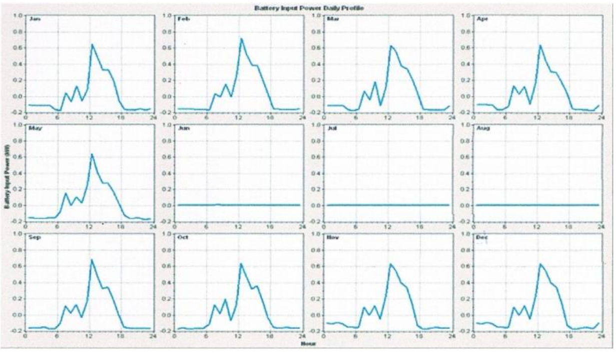


Fig. 16. Battery Park charge and discharge profile for a year

VI. Economic Study

Economic and technical studies of our hybrid system involved costs in system hybridization: Generally two main types of costs involved in hybridization of diesel power plant with photovoltaic can be distinguished:

- The investment costs
- The operating costs

The investment costs are involved in the implementation phase of the project, while the operating costs occur after the commissioning of the facility for the duration of use.

In general, operating costs of various system components are irregularly involved during the facility term use.

VI.1. Investment Cost on a Hybridization Project

The project investment cost, early involved in implementation phase takes into account all the costs related to the purchase of necessary components, to ensure transport, installation and system commissioning.

However, if the project can recover one or more generators, the cost of these groups will be considered a grain and will be deducted from the investment. In general, the investment costs depend on:

- Dimension and number of system components;
- The type and the different components;
- Costs of transport;
- Cost of installation and implementation.

VI.2. Cost Optimization

HOMER simulates all the possible combinations that are given in search space of each configuration and component. It also considers the sensitive variables which are an added advantage for obtaining nest configuration. Thus choosing the best feasible configuration and components with minimum net present cost is an optimization problem. This optimization problem is solved in HOMER based on Grahams Algorithm.

The output gives a standalone hybrid energy system considering all the factors like economics, reliability and metrological conditions subject to physical and operational constraints.

The Net present cost of the system is calculated by:

$$C_{NPc} = \frac{C_{ann,tot}}{CRF\left(i, R_{proj}\right)} \tag{8}$$

where $C_{ann,tot}$ is the total annualized cost, ‘i’ the annual interest rate, R_{Proj} the project life time and $CRF(i,N)$ is the capital recovery factor given by the equation:

$$CRF(i,N) = \frac{i(1+i)^N}{[(1+i)^N - 1]} \tag{9}$$

where N is number of years.

The equation to calculate the levelized cost of energy (COE) is:

$$COE = \frac{C_{ann,tot}}{E_{prim}} \tag{10}$$

where E_{prim} is the total amount of primary load.

Homer displays all the feasible solutions considering all the sources like Diesel,PV diesel battery which are explained in Table IV. Table IV includes all the costs of the hybrid system, the overall cost of the system being displayed to 29,390 $.

VI.3. Cost of Operation in a Hybridization Project

Operating costs in hybridization project are much more difficult to determine than investment costs. Investment costs occur on the date of payment of the investment so they occur simultaneously at the date of system commissioning. Thus, they are well known for the design phase and project design. The operating costs, however, will occur more or less regularly throughout the use phase from the date of system commissioning. They therefore contribute to future cash flows.

TABLE IV
INCLUDES ALL THE COSTS OF THE HYBRID SYSTEM

Component	Capital ($)	Replace -ment ($)	O&M ($)	Fuel ($)	Salvage ($)	Total ($)
PV	5.700	655	10.866	0	-367	16.854
Generator1	3.000	0	6	268	-629	2.646
Hoppecke 120PzS150	3.700	1.154	3.452	0	-647	7.659
Converter	1.600	668	89	0	-124	2.233
System	14.000	2.476	14.413	268	-1767	29.390

Accordingly, it is naturally estimated and predicted that some costs are inherent.

There are three main types of operating costs:

- Replacement costs (n Remple comp k)
- Maintenance costs and maintenance (cost has emhyb)
- Energy costs (cost to fuel eco)

VI.3.1. Total Annual Cost of the Project

The annual cost of the project is given by the formula: Cost is the total project cost has inv = + project cost has repl + system has cost hybem - has cost fuel eco.

VI.3.2. Simulation Results

To study the project's profitability it is possible to make a hybrid system simulation operation over 25 years.

VI.3.3. Project Operating Costs

The following chart shows the operating costs of each element of the system for 25years.

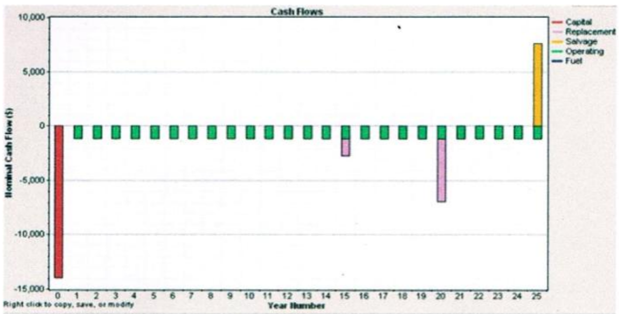


Fig. 17. Operating cost over 25 years

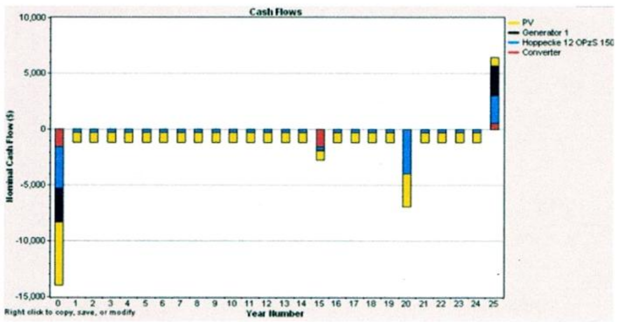


Fig. 18. Detailed profile of each cost up to 25 years

VI.3.4. Total Cost Of The Project

The following chart shows the breakdown of costs according to the different components of the HEU. In yellow is the cost of the PV generator followed by the cost of GD in Black. They are followed by the cost of batteries (blue) and converter (red).

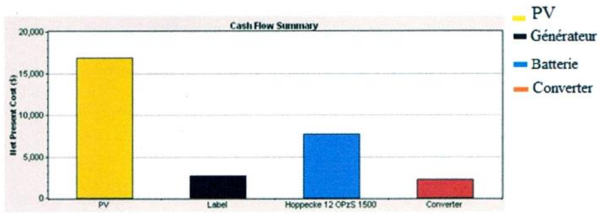


Fig. 19. Distribution of total project cost

PV generator cost is evidently much higher than other components, and this is justified by the distribution of electricity production by PV.

VII. Conclusion

In this work, a study with Homer software has been done to optimise an hybrid power system, combinate two energy resources,one renewable created by PV, and one non renewable generated by a fuel generator. This system is provided to a school in Tarat area located in south Algeria. The considered daily primary load is 4.8 kWh, with a peak of nominal power of 1.5kW. Since Wind speed is very low in Tarat, wind energy resources were not proposed. Also, wind turbine's price is very expensive and provides low energy production, less than solar sources production. Using Homer software, an optimal solution for the proposed hybrid system was obtained, and it was also shown that the cost of PV is 57,35% of the total hybrid system cost, and this system produces 96% of all hybrid system production, but the generator produces just 4%.

In four months (Jun, July, August and October), PV production was 100%, which texplains the benefits of the proposed system. In addition, fuel cost is sheeper in Algeria. This PV production percentage was never found in another site; compared to[20], in off grid hybrid system study cases in India; PV presents 28% with batteries storage; and 64% without storage. Another study in Bangladesh site [21], PV presents 89% from an hybrid power system. Therefore, the proposed site can give good benefits with a 96% and 100% production.

This study considers a power hybrid system off grid and PV without tracking, which will be taken into account in a future research project.

References

[1] A.N Supriyadiet T. Hashiguchi, "Control Scheme of Hybrid Wind-Diesel Power Generation System," *From turbine for wind farms*, JaponIndonisia, April 2011.

[2] R. Sebastiin, M. Castro, E. Sancristobal, F. Yeves and J. Peire, "Approaching hybrid wind-diesel systems and Controller Area

Network," *IECON 02, IEEE Condefrence*, pp- 2300-2305, Nov 2002
[3] D. Rekioua, Z. Roumila et T. Rekioua, "*Etude d'une centrale hybride photovoltaïque - éolien - diesel*," Université A. Mira, Bejaia. Bejaia, Algérie 2008.
[4] Nahid-Al-Masood, RifatMirza, Jubaer Ahmed, "Design of a Cost Effective Off-Grid Wind-Diesel Hybrid Power System in an Island of Bangladesh," *World Academy of Science, Engineering and Technology*, 2011.
[5] P.A. Stott and M.A. Mueller, "Modelling fully variable speed hybrid wind diesel systems," *UPEC'06*, Vol 1, pp-212-216, Newcastle USA, Sep 2006.
[6] Mohammad-Ali Yazdanpanah, "Modeling and sizing optimization of hybrid photovoltaic/wind power generation system," *J Ind Eng Int* (2014) 10:49.
[7] S.S.Murthy, S. Mishra, G. Mallesham and P. C. Sekhar, "Voltage and Frequency control of wind diesel hybrid system with variable speed wind turbine," *IEEE magazine* 20-23 Decembre 2010.
[8] P. S. Panickar, S. Rahman, S. M. Islam and T. L. Pryor, "*Adaptive control strategies in wind-diesel hybrid systems*," Curtin University, Western Australia.
[9] N. Vania and V. Kharea, "Rural Electrification System based on Hybrid Energy System Model Optimization using HOMER," *CJBAS* Vol. (01) - August - Issue 01 (2013) 19-25.
[10] R. Huang, S. H. Low, U. TopcuandK. Mani Chandy, "Optimal Design of Hybrid Energy System with PV/ Wind Turbine/ Storage: A Case Study," *Virtual Power Plants, Distributed Generation, Microgrids, Renewables and Storage Internationa IEEE conference*, 17-20 Oct. 2011.
[11] M. A. Matinn, A.Deb, A.Nasir, "Optimum Planning of Hybrid Energy System using HOMER for Rural Electrification," *International Journal of Computer Applications* (0975 – 8887) Volume 66– No.13, March 2013.
[12] Sheilla, M., Tan, C., Lim, C., Techno-Economic Analysis of a Photovoltaic-Fuel Cell Grid-Connected Hybrid Energy System, (2014) *International Review on Modelling and Simulations (IREMOS)*, 7 (1), pp. 65-75.
[13] Mohammed, O., Amirat, Y., Benbouzid, M., Feld, G., Tang, T., Elbaset, A.,Optimal Design of a Stand-Alone Hybrid PV/Fuel Cell Power System for the City of Brest in France, (2014) *International Journal on Energy Conversion (IRECON)*, 2 (1), pp. 1-7.
[14] Mohamed, R., Ibrahim, D., Youssef, H., Rakha, H., Optimal Sizing and Economic Analysis of Different Configurations of Photovoltaic systems, (2014) *International Review of Electrical Engineering (IREE)*, 9 (1), pp. 146-156.
[15] Gan, C., Prayun, W., Tan, C., Shamshiri, M., Design of a Hybrid Diesel/PV/Wind/Battery System in Remote Areas, (2014) *International Review of Electrical Engineering (IREE)*, 9 (2), pp. 420-430.
[16] [12] Acakpovi, A., Hagan, E. B. 2013. Novel Photovoltaic Module Modeling using Matlab/Simulink, *International Journal of Computer Applications (IJCA)* Vol. 83, No.16, pp 27-32.
[17] Villalva, M., G., Gazoli, J.R., and Filho. E.R. 2009. Comprehensive Approach of Modeling and Simulation of a Photovoltaic Arrays. *IEEE Transaction on Power Electronics.* Vol.24. No.5. pp. 1198-1208.
[18] Ramos-Paja, C.A., Perez, E., Montoya, D.G., Carrejo, C.E., Simon-Muela, A., Alonso, C. 2010. *Modelling of Full Photovoltaic Systems applied to Advanced Control Strategies*. Columbia: Universidad Nacional de Columbia
[19] Tsai, H.L., Tu, C.S., and Su, Y.J. 2008. Development of Generalized Phottovoltaic Model Using MATLAB/SIMULINK. *Proceedings on the world congress on Engineering and Computer Science. WCECS*, ISBN: 978-988-98671-0-2, 6p
[20] Charanjiv Gupta, Sanjeev Singh, Hybrid power systems- a comparative analysis, *Journal of Scientific Research and Development*, 3 (4) 2016, Pages: 55-59
[21] Saikat Roy, M. M. Rhaman "Hybrid Solar Power Plant in Saint Martin's Island can Enlarge Tourist Attraction in Bangladesh", *I.J. Engineering and Manufacturing*, 2016, 3, 12-22 Published Online May 2016 in MECS.
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